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Building capacity, strengthening knowledge foundations and supporting policy

Data and knowledge management policy and code of practice on the use of artificial intelligence

Note by the secretariat

1. The present information note sets out the outcomes of the work of the task force on data and knowledge management and its dedicated technical support unit during the intersessional period 2024–2026, conducted in support of objective 3 (a) of the rolling work programme of the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services (IPBES) up to 2030, on advanced work on knowledge and data.

2. In response to decision IPBES-11/1, the task force prepared a revised version of the IPBES data and knowledge management policy (version 2.2) and an IPBES code of practice on the use of artificial intelligence, tailored to the needs of assessments. The texts are set out, without formal editing, in the annex to the present note.

3. The documents have been reviewed by the task force on data and knowledge management, the Multidisciplinary Expert Panel and the Bureau.

* IPBES/12/1.

Annex*

Report on the outcomes of the activities conducted during the intersessional period 2024–2026 related to the IPBES data and knowledge management policy, the IPBES long-term vision on data and knowledge management, and the code of practice on the use of artificial intelligence, tailored to the needs of the assessments

I. Review and revision of the IPBES data and knowledge management policy

1. Following Plenary decision IPBES-11/1, the task force on data and knowledge management prepared a revised draft of the IPBES data and knowledge management policy (version 2.2). The revision built upon version 2.1, adopted at IPBES-11 (December 2024), and updates were introduced to maintain alignment with ongoing developments across the Platform and with broader principles and guidance relevant to ethical and responsible data governance.

2. The revisions align the policy with the principles and guidance of the IPBES code of practice on the use of artificial intelligence, tailored to the needs of assessments (AI code of practice), particularly with respect to the ethical management, transparency and accountability of data and knowledge products supported by IPBES. The updates also reinforce the application of the FAIR and CARE principles and strengthen provisions on accessibility, inclusivity and collective benefit. Major updates include:

(a) Revision of the roles and responsibilities of the technical support units of the assessments to incorporate three priority areas previously emphasized by the Multidisciplinary Expert Panel and the Bureau and consistent with the AI code of practice;

(b) A streamlined structure of the document, maintaining a concise and high-level focus while ensuring that operational details can be updated more flexibly in supporting annexes; and

(c) Revision of the glossary to enhance clarity, consistency and alignment with terminology used in the AI code of practice.

3. The revised policy specifies that any use of advances in data technologies, including artificial intelligence tools, within IPBES work should be transparent and well documented. Such uses are to be indicated during the milestone reviews of assessment drafts, with information on their purpose, context and verification to ensure traceability and accountability.

4. The revised policy was submitted to the Multidisciplinary Expert Panel and the Bureau for review at their twenty-fourth meeting in July 2025. The version set out in annex II incorporates comments and suggestions received from the task force, the Multidisciplinary Expert Panel and the Bureau, and the secretariat following their review.

5. In addition, a selected group of external experts, including former members of the task force and specialists identified through the experience and professional networks of the technical support unit, was invited to review sections related to the ethical use of artificial intelligence, transparency and data responsibility, complementing the internal review process.

6. The revision process was coordinated by the technical support unit for data and knowledge management under the guidance of the task force. It included several ad hoc meetings and exchanges and steps to maintain consistency with the AI code of practice and the long-term vision for data and knowledge management, thereby supporting the coherence of the overall IPBES framework for data and knowledge governance.

7. Version 2.2 of the policy on data and knowledge management is set out in appendix I to the present document.

* The annex has not been formally edited.

II. Development of a code of practice on the use of artificial intelligence, tailored to the needs of the assessments

8. In accordance with Plenary decision IPBES-11/1, the task force on data and knowledge management developed the IPBES code of practice on the use of artificial intelligence, tailored to the needs of assessments. The code provides practical guidance on the use of artificial intelligence within IPBES processes and supports the implementation of the IPBES data and knowledge management policy.

9. As part of its preparation, the technical support unit carried out a review of existing guidelines, codes of practice and technical notes on the application of artificial intelligence across United Nations organizations, conventions, multilateral environmental agreements and other intergovernmental science-policy platforms, as well as selected regional and national directives. The review also considered recent international and regional frameworks, including *Governing AI for Humanity – Final Report of the United Nations High-Level Advisory Body on Artificial Intelligence*[1], the Organisation for Economic Co-operation and Development report *Biodiversity and AI: Opportunities and Recommendations for Action*[2], and the European Union Artificial Intelligence Act[3]. The code of practice is consistent with the UNESCO Recommendation on the Ethics of Artificial Intelligence, the Principles for the Ethical Use of Artificial Intelligence in the United Nations System[4], and relevant IPBES policies and procedures.

10. The code of practice outlines general principles and specific use cases that reflect common and anticipated applications of artificial intelligence in the preparation of assessments. It covers editorial and translational usage, coding support, literature searches, data and text analysis, figure preparation and the review of drafts. For each use case, it provides practical instructions on responsible use, associated risks and limitations, and the documentation needed to support transparency and reproducibility. The code specifies that all outputs derived from or influenced by artificial intelligence remain the responsibility of authors and that the confidentiality of draft materials is to be maintained throughout the assessment process.

11. The draft code of practice was submitted to the Multidisciplinary Expert Panel and the Bureau for consideration at their twenty-fourth meeting in July 2025. Version 1.0 incorporates comments and suggestions received from the task force, the Multidisciplinary Expert Panel, the Bureau and the secretariat following their review.

12. In addition, a selected group of external experts, including former members of the task force and other specialists, was invited to provide feedback on the draft. Their review included attention to ethical safeguards, reproducibility and the practical applicability of the code within assessment workflows.

13. The IPBES code of practice on the use of artificial intelligence, tailored to the needs of assessments (version 1.0), is set out in appendix II to the present document.

III. Review and revision of the IPBES long-term vision of data and knowledge management

14. Following Plenary decision IPBES-11/1, the task force on data and knowledge management reviewed the IPBES long-term vision on data and knowledge management (version 1.0). The review was coordinated by the technical support unit and guided by the discussions of the task force during the current intersessional period.

15. The objective of the review was to assess the alignment of the long-term vision with the IPBES code of practice on the use of artificial intelligence, tailored to the needs of assessments, and to consider the implications of recent progress and policy developments for the roadmap targets up to 2030. The review reaffirmed the importance of embedding the FAIR and CARE principles, ensuring responsible and transparent use of artificial intelligence, and strengthening traceability, accessibility and interoperability across IPBES knowledge products.

[1] https://www.un.org/sites/un2.un.org/files/governing_ai_for_humanity_final_report_en.pdf.
[2] https://wp.oecd.ai/app/uploads/2025/05/biodiversity-and-AI-opportunities-recommendations-for-action.pdf.
[3] http://data.europa.eu/eli/reg/2024/1689/oj.
[4] https://www.unesco.org/en/articles/recommendation-ethics-artificial-intelligence.

16. The long-term vision continues to provide a high-level framework for future developments in data and knowledge management within IPBES. It provides a coherent structure to guide actions up to 2030 and will inform the development of a strategic roadmap in the next intersessional period.

Appendix I

The IPBES Data and Knowledge Management Policy

Version: 2.2[5]

February 2026

Introduction

This policy builds on the data and information management plan approved by the IPBES Plenary set out in annex II to decision IPBES-3/1, and the earlier revision of the IPBES data and knowledge management policy welcomed by the IPBES Plenary at its ninth session, set out in IPBES/9/INF/14, appendix II to the annex, and the most recent version outlined in IPBES/11/INF/26. It builds specifically on the activity "reviewing and developing data and metadata guidelines", and is grounded in its principles for "managing knowledge, information, and data" in the Platform, in particular open science and accessibility. Definitions of the terms used throughout this document can be found in annex A.

- **Open science.** The open science approach promotes the generation of knowledge through collaboration based on free and open access to knowledge, information, and data. Open science, therefore, ensures that the work of all the IPBES experts and stakeholders involved in the development of an IPBES product is fully recognised, properly attributed, documented, and preserved. Adoption of the open science approach and principles means a significant cultural change in the way science-policy is conducted, and the outcome of knowledge synthesis and the underlying workflows are shared publicly. In the context of the Platform, the open science approach brings very significant advances in transparency, data integration, analysis and interpretation and could ultimately lead to a better understanding of nature and its contribution to human well-being.
- **Accessibility**. A core value of the Platform is free and open access to its deliverables and to the material on which they are based. Consequently, the policy will aim for open, sustained access to data and information sources for its deliverables (e.g., in the scientific literature) with as few restrictions as possible; enforce the use of common and accessible, non-proprietary file formats in the Platform's deliverables; emphasise the need to make data and knowledge available; and facilitate multilingual discovery and sharing of data and knowledge wherever possible. The Platform acknowledges that making data and information accessible may not always mean it is openly and freely accessible to all IPBES members and stakeholders; this includes considerations with regard to Indigenous and local knowledge (ILK) and Indigenous Peoples and local communities (IPLCs), as technical, economic, political or other reasons may limit accessibility, but not its findability.
- **Inclusivity and collective benefit**. Cooperation in knowledge synthesis and broad acceptance of the resulting IPBES products is essential for the Platform to fulfil its mandate. For the acceptance and cooperation of stakeholders and data and knowledge holders, inclusivity in all stages of the knowledge synthesis is essential, and IPBES takes many steps to try to enhance participation with all stakeholders, including IPLCs. IPBES acknowledges the richness of diverse knowledge systems, including ILK, and the diversity of knowledge holders and producers, including IPLCs.
- **Responsible and ethical use of advances in data technologies**. The ethical and responsible use of advances in data technologies, in particular generative artificial intelligence (AI), ensures the appropriate use of these technologies and fosters trust in the Platform. The use of AI should be grounded in transparency, accountability, and acknowledgement of potential biases. The values of inclusivity, accessibility, and collective benefit should be considered in the use of AI. This approach supports the equitable advancement of knowledge and ensures that the use of AI benefits IPBES products in accordance with this policy and the Code of Practice on the ethical and responsible use of AI.

Thus, this policy follows, regarding data and knowledge management within IPBES, the overarching strategies of UNEP, policies and guidelines, to the maximum extent possible within the mandate of the

[5] Suggested citation: IPBES (2026): IPBES Data and Knowledge Management Policy, version 2.2. Krug, R. M. and Niamir, A. (eds.). IPBES secretariat, Bonn, Germany. DOI: 10.5281/zenodo.3551078

Platform, allowing IPBES members and stakeholders, including IPLCs and other interested parties, to use and access IPBES products, and data collected or documented during their production and consequently derive benefit from them. IPBES has recognised the importance of ILK for the conservation and sustainable use of ecosystems, and detailed procedures and protocols to be used with regard to ILK and the engagement with IPLCs have been developed. From its inception, the Platform aims to enhance interrelationships and complementarities between different knowledge systems. By promoting transparency, inclusivity, and the ethical use of data and technologies, this policy contributes to building and maintaining trust among all knowledge holders and users.

Objectives

To fulfil its function to produce scientifically robust and transparent assessments, IPBES is committed to implementing data and knowledge management procedures that are discipline-appropriate, practical, cost-effective, sustainable, and supportive of its objectives. This data and knowledge management policy is the primary reference document for IPBES data and knowledge management. It serves to ensure that data and knowledge are managed correctly and consistently, are maintained to the highest possible standard, and contribute to building trust through transparent processes.

The data and knowledge management policy has the following objectives:

a. To ensure that data and knowledge produced during IPBES knowledge synthesis activities follow legal obligations and the FAIR data principles (Findable, Accessible, Interoperable and Reusable) and CARE principles for Indigenous data governance (Collective benefit, Authority to control, Responsibility and Ethics) to the fullest extent possible within the mandate of the Platform;

b. To ensure that IPBES products, to the maximum extent possible within the mandate of the Platform, are openly available and designed so they are accessible, allowing all scientists, IPBES members, IPLCs and other stakeholders to use them and consequently derive and share the benefits from them, thereby fostering transparency and trust;

c. To give full consideration the use of advances in data technology, in particular AI, based on existing and evolving principles and guidelines for the applicability and ethical and responsible use;

d. To provide guidance to all IPBES entities, including the secretariat and IPBES experts, to fulfil their responsibilities with respect to the management, handling, preservation and distribution of data and knowledge, generated data, and IPBES products within the Platform;

e. To provide guidance for the long-term storage and preservation of IPBES products to the IPBES experts;

f. To promote the usage of open-source software to enable users to recreate and use IPBES products without limitations;

g. To guide the IPBES experts to fulfil their responsibilities to develop the necessary data and knowledge management reports (DMRs), which fulfil the requirements of this policy and support consistency, transparency and accountability in data practices.

General principles

IPBES products and the associated knowledge synthesis should be managed following the FAIR and CARE principles to the fullest extent possible within the mandate of the Platform throughout their life cycle, in line with the commitment to open science and accessibility, as these are essential for fulfilling the functions of IPBES: to perform regular and timely assessments of knowledge on nature, its contributions to a good quality of life and their interlinkages, in a transparent and reproducible manner. IPBES products and associated knowledge synthesis relating to IPLCs or their knowledge should follow the IPBES approach to recognising and working with ILK (annex II to decision IPBES-5/1).

In the management, handling and delivery of IPBES products, international, regional and national law should be complied with, which includes access and benefit-sharing, sovereign rights over genetic resources, rights of privacy, intellectual property rights, data governance regulations and confidentiality duties. If necessary, IPBES products should be anonymised before long-term storage and publication.

IPBES experts are encouraged to take advantage of advances in data technologies and to make best use of these digital tools throughout the assessment process and in the preparation of IPBES products.

These technologies should be used responsibly, with particular attention to ethics, transparency and underlying biases. The use of AI within IPBES should be precautionary, following the Principles for the Ethical Use of Artificial Intelligence in the United Nations System and the UNESCO Recommendation on the Ethics of Artificial Intelligence. IPBES provides guidance to all IPBES experts to ensure that they are aware of and follow IPBES procedures.

Application

This data and knowledge management policy will be applied to all new and on-going IPBES products and related knowledge synthesis. The policy should be reviewed at least every two years by the task force on data and knowledge management to adapt to new developments in these principles, taking full account of existing and evolving principles and guidelines on the applicability and ethical use of advances in data technology. Exceptions and deviations from this policy will be reviewed by the technical support unit on data and knowledge management and the secretariat, and submitted for approval to the Management Committee, and documented in a corresponding data management report (DMR).

Scope

This policy applies to all IPBES products and knowledge synthesis. IPBES experts are required to abide by the terms and conditions agreed with third parties. The current IPBES data and knowledge management policy also recognises that the policies of third parties are evolving and that they may require higher levels of data accessibility and dissemination in the future.

Compliance and enforcement

Compliance with the data and knowledge management policy is mandatory for all IPBES entities involved in the preparation of IPBES products. Compliance will be monitored by the technical support unit on data and knowledge management. Products will not be accepted as IPBES products unless they comply with this policy. The roles and responsibilities for the implementation of this policy and the production of IPBES products compliant with all IPBES data-related tasks are distributed among all IPBES entities according to their roles. The detailed roles and responsibilities are outlined in annex B.

Provisions on data and knowledge management reporting

a. DMR is expected for each IPBES task. This can be achieved by a single DMR covering all the analyses and work for an entire IPBES task, or by separate DMRs for each subtask, such as an analysis or the development of one figure, within an IPBES task.

b. DMR can consist of the report and, optionally, multiple additional documents. When it contains multiple documents, the actual DMR — the main document — can be considered an overview document that provides transparency on the methods, with the additional documents providing more technical background for reproducibility.

c. All activities and outputs related to data and knowledge management must comply with this policy. If full compliance is not possible within assessment related tasks, the responsible experts and the relevant assessment technical support unit must clearly specify and justify any exceptions in the DMR. Such exceptions will be reviewed by the technical support unit on data and knowledge management and the secretariat, and submitted for approval to the Management Committee.

d. IPBES experts are responsible for ensuring that DMRs are created, maintained and updated throughout the life cycle of each task and submitted to the relevant assessment technical support unit in time for the next milestone. The technical support units, both for assessments and for data and knowledge management, will provide guidance, templates and support to assist experts in the preparation and maintenance of DMRs, ensuring consistency with this policy.

e. The technical support unit on data and knowledge management provides support and, where appropriate, guidelines and examples to the IPBES experts to make sure that FAIR and CARE principles are followed to the fullest extent possible within the mandate of the Platform for data and knowledge management and documentation in the DMRs.

f. The task force and the technical support unit on ILK provide support and, where appropriate, guidelines and examples to the IPBES experts to make sure that the FAIR and CARE principles are followed to the fullest extent possible within the mandate of the Platform for data and knowledge management and documentation in the DMRs, where such data relate to ILK or IPLCs.

g. The secretariat provides information about recommended long-term, and to the extent possible certified, open data repositories that provide digital object identifiers (DOIs).

Provisions on accessibility, inclusivity and collective benefit

a. IPBES products should be preserved, including a DOI for each milestone of a task.

b. IPBES products, including those in or associated with an assessment such as DMRs, should be made openly accessible in a form that follows this policy at the latest six months after the approval or acceptance of the assessment, or approval or acknowledgement of other IPBES products by the Plenary. Data and knowledge related to milestones should also be made accessible, as far as confidentiality rules allow. Embargo periods are possible but need to be communicated with the secretariat in advance. Restricted access to the IPBES data underlying IPBES products is mandatory until approval of the assessment by the Plenary, with the exception of the external review periods. After approval, restricted access is only allowed under special circumstances and needs to be assessed by the secretariat and approved by the Management Committee and communicated to the secretariat.

c. Applicable ethical, privacy, confidentiality and data governance requirements need to be followed and generated data, if deemed necessary, must be anonymised before preservation.

d. Tools for the finding, processing and final presentation of any data should be used in a responsible and ethical manner that follows this policy. No tools should be used without critical assessment of the methods and results by the experts.

e. The management, handling and delivery of materials from IPLCs adhere to the FAIR and CARE principles to the fullest extent possible as outlined in this policy, as well as to other binding conditions outside this policy in accordance with international, regional and national law.

f. IPBES products and their metadata should be released with a clear and accessible licence. Allowed licences for IPBES products are Creative Commons By Attribution (CC-BY) or licences equivalent to these, which permit users to copy, distribute and transmit work, and adapt work under the condition that the user must attribute the work in the manner specified by IPBES. Divergent licences need to be approved by the technical support unit on data and knowledge management.

g. All knowledge synthesis within IPBES has to be conducted in accordance with agreements with the holders of the data and knowledge regarding use, reuse, presentation and procedures. Communication with IPLCs needs to be maintained over the whole task cycle to the extent possible within the mandate of the Platform to provide feedback to the knowledge holders and include feedback from the knowledge holders in the knowledge synthesis, as outlined in the IPBES approach to recognising and working with ILK (annex II to decision IPBES-5/1).

h. IPBES products will be made available and accessible to the holders of the data and knowledge, in line with agreed terms documented during ILK dialogue workshops or other activities, with due consideration to the FAIR and CARE principles.

Provisions on the use of advances in data technologies

a. IPBES experts are encouraged to make use of advances in data technology, provided their use is compliant with this policy and applicable IPBES codes of practice, and follows a precautionary approach.

b. AI systems may introduce risks, particularly in the interpretation and application of results, which require careful consideration to prevent negative consequences.

c. The use of AI in IPBES tasks, where permitted, must remain transparent, with well-documented prompts.

d. Further guidance and examples of ethical and responsible AI use are provided in the IPBES code of practice on the use of artificial intelligence, tailored to the needs of assessments. This includes alignment with the FAIR and CARE principles to ensure that AI systems contribute to just, transparent and inclusive knowledge synthesis.

Annex A: Glossary

Access and benefit-sharing. Rooted in the CARE principles, this refers to the understanding that data, knowledge or resources should be used in ways that benefit the community or group from which they originate. This principle ensures that the interests of the community are prioritised and that any use of data or resources contributes positively to their well-being and development.

Accessible. Refers to the ease with which data, knowledge and IPBES products can be found, retrieved and used by all intended audiences, including IPBES members, IPLCs and other stakeholders. Accessibility implies that information is available in formats and languages that are understandable and usable, with minimal technical, legal or financial barriers, and that support discoverability and inclusiveness.

AI (artificial intelligence). AI models **are** trained with large amounts of data. These models and prompts can be embedded in interfaces that resemble familiar tools (for example, literature search engines or grammar checkers). In this document, AI refers to all tools that directly interact with generative AI (for example, chatbots) or use AI in the background. The term is limited to generative AI systems.

CARE. A set of guiding principles for Indigenous data governance, focusing on the appropriate use and reuse of Indigenous data and knowledge. CARE stands for Collective Benefit, Authority to Control, Responsibility and Ethics. These principles should also be applied to the management of knowledge.

Citations and references. A citation refers to the source of information supporting IPBES deliverables, addressing where the information came from. A reference includes sufficient details to make the source findable and traceable.

Collective benefit. Benefits for all. In the context of this policy, these are direct benefits resulting from the use of and access to IPBES products, and knowledge and data gathered or documented during their production and from sharing credit. To this end, IPBES products and associated knowledge and data should, to the maximum extent possible, be openly available and designed so they are accessible to all, allowing scientists, IPLCs and others to use them and derive benefit from them.

Data. Data can be of any nature: spatial or non-spatial, qualitative or quantitative, descriptive, and from all scientific disciplines. This includes information from IPLCs.

Data and knowledge. Data and knowledge often form a continuum, where data cannot be interpreted without knowledge. They include individual units of information obtained from observations, measurements, experiences or value systems, and form the basis for monitoring, synthesis, assessments and analysis.

Data deposit package. The content deposited in a long-term repository. Each package has a DOI and consists of the data and knowledge and a DMR describing them.

DMR (data and knowledge management report). A document that forms part of each IPBES assessment and associated tasks, detailing how data and knowledge are handled throughout the process. It includes information on data creation, references, scripts, software used and access provisions, and must be updated throughout the duration of the task or assessment.

DOI (digital object identifier). A digital identifier for a physical or digital object, defined in the DOI Handbook and recognised under ISO 26324:2012. Provides a persistent and interoperable link.

Ethical and responsible use. The ethical and responsible use of AI involves ensuring that its development and application uphold human rights, dignity and fairness, and that AI systems contribute to sustainable development and the well-being of people and the planet. It requires transparency, accountability, inclusiveness, reliability, safety and respect for privacy.

External data and knowledge. Data and knowledge generated outside IPBES and published in peer-reviewed journals, grey literature or available as ILK. While IPBES is not responsible for preserving these, it must preserve metadata describing the data's provenance, licensing and copyright information when such knowledge is used, with agreement from knowledge holders.

FAIR. A set of guiding principles for making data findable, accessible, interoperable and reusable. These should also be applied to knowledge management.

Generative AI. AI that produces content such as images, code or text in response to user input. Well-known examples include chatbots.

ILK (Indigenous and local knowledge). As defined in the IPBES assessment glossaries, Indigenous and local knowledge systems are "social and ecological knowledge, practices and beliefs concerning relationships among living beings and their environments." See also annex II to decision IPBES-5/1, which describes them as "dynamic bodies of integrated, holistic, social and ecological knowledge, practices and beliefs pertaining to the relationship of living beings, including people, with one another and with their environments."

Knowledge. Understanding developed through experience, reasoning and learning. Essential for interpreting associated data.

Knowledge synthesis. All IPBES activities that collect, measure, integrate or analyse data or knowledge, including ILK. Also includes documentation of ILK from dialogue workshops or IPLC activities.

Milestone. A significant step in an IPBES task or assessment that warrants long-term storage. For assessments, this includes all drafts prepared for internal and external reviews as well as final versions of chapters.

Workflow. A repeatable set of steps for achieving a goal, such as data gathering, analysis and visualisation. Workflows document the scientific process, including tools and software used.

Annex B: Roles and responsibilities

Bureau and MEP

- review any changes to the policy as proposed by the task force on data and knowledge, and consider these for approval.

Secretariat

- execute, under the guidance of the task force on data and knowledge and in cooperation with the technical support unit (TSU) on data and knowledge management, the development and maintenance of guidelines, tutorials, workflows and examples to enable IPBES experts to implement this policy.
- maintain an accurate, up-to-date and accessible list of citations and references (including rich metadata), as well as links to external data and knowledge, and knowledge and generated data used in IPBES products.

Task force on data and knowledge management

- develop and provide guidelines and examples for data and knowledge management, and guide their development and maintenance.
- advise the TSU on data and knowledge management.
- review the policy at least every two years.
- determine procedures that comply with legal requirements and fulfil the FAIR and CARE principles, including through collaboration with the ILK task force, to the fullest extent possible within the Platform's mandate.
- review the annexes, guidelines, tutorials, workflows and examples related to this policy annually to identify gaps and implement new developments.

Task force on ILK

- work with the task force on data and knowledge to review the policy, at least every two years, with regard to aspects relating to IPLCs and ILK, including determining procedures that fulfil FAIR and CARE principles to the fullest extent possible within the Platform's mandate.
- review the guidelines, tutorials, workflows and examples related to this policy, particularly with regard to aspects relating to IPLCs and ILK, on a yearly basis to identify gaps and implement new developments.

TSU on data and knowledge management

- provide support, advice and participate in efforts from the task force on data and knowledge management, and execute, under guidance and in cooperation with the secretariat, the development and maintenance of guidelines, tutorials, workflows and examples to enable IPBES to implement this policy.

- review the DMRs from corresponding assessment TSUs to ensure alignment with the data and knowledge management policy.
- ensure that assessment TSUs follow the data and knowledge management policy, fulfil their responsibilities as outlined in this policy and in the DMRs, and will collect metadata of the IPBES products from the TSUs to ensure accessibility and discoverability.
- provide guidance to IPBES experts on the responsible use of advances in data technologies.

TSU on Indigenous and local knowledge

- provide support, advice and participate in efforts from the task force on data and knowledge management to develop guidelines and examples for data and knowledge management and reporting when working with ILK and in alignment with CARE principles.
- support the implementation of these guidelines in IPBES assessments and other processes.
- review DMRs upon request from the TSU on data and knowledge management to ensure alignment with ILK and IPLC considerations.

Assessment TSU

- collect the DMRs from IPBES experts involved in assessments.
- assist experts in ensuring DMRs adhere to this policy.
- assign a persistent identifier (for example, DOI) for each data deposit package.
- add consistent, specific, and descriptive keywords and metadata (for example, chapter, assessment, figure) to make data findable and identifiable.
- ensure that IPBES experts fulfil responsibilities as outlined in this policy and the DMRs, and will collect product metadata for transfer to the task force on data and knowledge management.
- develop and maintain, under the guidance of the TSU on data and knowledge management and in cooperation with the secretariat, guidelines, tutorials, workflows and examples (DMRs) to enable implementation of this policy.
- check for retracted references in IPBES products, ensuring these are replaced prior to the final milestone.
- collect and provide feedback to the TSU on data and knowledge management on lessons learned during policy implementation.

IPBES experts

- prepare and keep DMRs up to date for all IPBES-related knowledge synthesis activities. DMRs should be available by the first milestone and updated for each subsequent milestone, conforming to this policy and the technical guidelines.
- responsible for implementing the DMRs and reporting to the relevant assessment TSU.
- responsible for the ethical, transparent and validated use of AI, particularly in line with responsible use principles.

Appendix II

The IPBES Code of Practice on the Use of Artificial Intelligence, Tailored to the Needs of Assessments

Version: 1.0[6]

February 2026

Introduction

This code of practice outlines principles for the ethical and responsible use of artificial intelligence (AI) in the preparation of IPBES deliverables, in particular assessments. It provides guidance for the use of AI tools in typical assessment activities, including the collection, analysis, processing, synthesis and presentation of data and knowledge in all formats, such as written content, figures and visualisations. The use of AI in the publication and dissemination of IPBES products is beyond the scope of this code. The code is organised around representative use cases that reflect common and emerging applications of AI in IPBES knowledge synthesis. These examples are not exhaustive but illustrate relevant risks, limitations and responsibilities associated with AI use.

AI technologies involving large language models and related systems are evolving rapidly. While they offer powerful capabilities to support experts, they also introduce specific challenges, including bias in training data and algorithms, uncertainty about copyright in training material and generated content, inconsistencies in AI-generated outputs and the risk of presenting inaccurate or misleading information as fact.

In line with the platform's data and knowledge management policy, including transparency, inclusivity, openness, recognition and respect for diverse knowledge systems, and the promotion of collective benefit, AI must be used in a manner that strengthens the credibility, legitimacy, integrity and inclusivity of IPBES products. Ethical and responsible use of AI, as defined in the UNESCO Recommendation on the Ethics of Artificial Intelligence (2021) and the UN Principles for the Ethical Use of Artificial Intelligence in the United Nations System (2022), requires transparency, accountability and fairness in all applications. Any use of AI tools in assessment preparation must therefore be documented in the corresponding data and knowledge management report to ensure traceability and accountability. Above all, responsible use of AI must maintain and foster trust among all stakeholders.

The development of this Code of Practice draws on relevant international and regional frameworks, including Governing AI for Humanity – Final Report of the United Nations High-Level Advisory Body on Artificial Intelligence[7], the Organisation for Economic Co-operation and Development (OECD) report on Biodiversity and AI: Opportunities and Recommendations for Action[8], and the European Union Artificial Intelligence Act[9]. The AI code of practice is aligned with the UNESCO Recommendation on the Ethics of Artificial Intelligence[10], the Principles for the Ethical Use of Artificial Intelligence in the United Nations System[11], and relevant IPBES policies and procedures.

General principles

The following general principles apply to all uses of artificial intelligence (AI) in the preparation of IPBES assessments. They are intended to ensure that AI is used ethically, transparently and in a way that supports the credibility, legitimacy, integrity and inclusivity of IPBES products.

Responsibility and accountability. All outputs derived from or influenced by AI must be critically reviewed and evaluated by IPBES experts. Responsibility for all content, including text, graphs, scripts, citations and references, lies with the authors. AI may not be cited as a source of

[6] Suggested citation: IPBES (2026): IPBES Code of Practice on the Use of Artificial Intelligence, Tailored to the Needs of Assessments, version 1.0. IPBES secretariat, Bonn, Germany. DOI: 10.5281/zenodo.17628864

[7] https://www.un.org/sites/un2.un.org/files/governing_ai_for_humanity_final_report_en.pdf

[8] https://wp.oecd.ai/app/uploads/2025/05/biodiversity-and-AI-opportunities-recommendations-for-action.pdf

[9] http://data.europa.eu/eli/reg/2024/1689/oj

[10] https://www.unesco.org/en/articles/recommendation-ethics-artificial-intelligence

[11] https://unsceb.org/principles-ethical-use-artificial-intelligence-united-nations-system

authority. Authors should be prepared to justify and defend the validity and appropriateness of AI-informed content in the same way as for any other assessment content.

Transparency and disclosure. Any use of AI must be disclosed and documented in the respective data and knowledge management report (DMR). Documentation should include the purpose of use, the tool(s) used, verbatim prompts or representative excerpts, the model type and configuration, and any relevant limitations or uncertainties. Where available, experts may refer to model cards, system cards or equivalent documentation describing model characteristics. Experts should also note whether they have trained or fine-tuned a model.

Prompt design and documentation. The formulation of prompts—including their structure, wording and level of detail—strongly influences AI-generated outputs. Experts should design prompts carefully to avoid introducing bias, inaccuracies or unverified assumptions. Prompts used at any stage of the assessment should be recorded in the DMR, together with a brief description of their purpose and outcome. When online tools are used, prompts must not include confidential IPBES text or identifiers.

Bias awareness and mitigation. Experts must remain vigilant to potential biases embedded in AI models through their design, training data or post-processing. AI-generated content must be critically reviewed to avoid reinforcing stereotypes, excluding certain perspectives or introducing misleading interpretations. When biases are identified, they should be corrected using alternative approaches or acknowledged and justified in the relevant text and in the DMR.

Scientific integrity and reproducibility. Because AI systems are stochastic and opaque, reproducibility may be limited. To maintain scientific integrity, experts should follow the above guidance on traceability, transparency and accountability, and ensure that AI-supported outputs are independently verifiable to the extent possible.

Respect for data governance and intellectual property. AI must not be used in ways that infringe or circumvent copyright, licensing or data-sharing agreements. As AI models may generate or reproduce copyrighted material, outputs should be checked using non-AI methods to ensure proper attribution and to avoid plagiarism, in accordance with applicable citation standards and the IPBES code of conduct for authors.

Alignment with United Nations core values. The use of AI should respect the United Nations values of integrity, professionalism and respect for diversity. Integrity requires the application of high standards of honesty and accountability; professionalism requires that AI is used responsibly and in line with the IPBES mandate; and respect for diversity requires inclusivity and fairness, avoiding discrimination or bias in AI-assisted processes.

Guided by the IPBES data and knowledge management policy. The use of AI in IPBES assessments should adhere to the principles set out in the IPBES data and knowledge management policy.

Confidentiality. The use of AI must comply with the IPBES code of practice for experts. Assessment drafts and related materials must remain confidential. Confidential information—including assessment text, lists of candidates, experts' personal data, external review comments and internal discussions—should only be processed using private, locally hosted or institutionally provided AI tools, and not shared with public or third-party commercial systems. When online tools are used, experts should not upload entire documents and must ensure that no confidential IPBES content or identifiers are included in prompts.

Specific use cases

This section provides practical guidance for the use of AI in the production of IPBES assessments. It aims to help experts minimise risks and ensure that all AI-related activities remain consistent with the IPBES data and knowledge management policy and the general principles outlined above. Each use case illustrates both opportunities and risks associated with AI applications. These examples are not exhaustive but are intended to support responsible decision making when integrating AI into the assessment workflow. Core assessment activities, including drafting assessment text, conclusions or findings, must not be carried out by AI systems.

Editorial usage

AI is increasingly embedded in writing and editing tools. These systems assist with improving spelling, grammar and clarity and may help align text with institutional styles such as the

United Nations Editorial Manual. Their use is common and often unavoidable in collaborative writing environments. AI-generated text must not be used directly in assessment drafts. Experts must independently write and verify all assessment content to ensure accuracy, neutrality and defensibility.

Spell checkers:

Usage: Continuous use during drafting and editing.

Risks: Minimal, though online tools may collect contextual data.

Guidelines: Spell checkers with AI imbedded in document drafting software. Experts should only use other AI spell checkers if they are local or offline tools. When using online tools, only short text segments should be processed. But no draft text from the assessment should be uploaded on AI tools. No IPBES or assessment content or identifiers should be included, in line with the principle of confidentiality. All corrections must be manually verified.

Grammar checkers:

Usage: During drafting and final editing.

Risks: Potential distortion of meaning; disclosure of text to external servers.

Guidelines: Experts may use AI grammar tools with the same caution as spell checkers. Suggestions must always be manually reviewed to ensure accuracy and preserve the intended meaning and tone.

Writing support:

Usage: Throughout drafting or when finalising text.

Risks: Confidentiality concerns; unintentional plagiarism; factual inaccuracies; homogenisation of style; distortion of intended meaning.

Guidelines: Experts may use AI tools for writing support, preferably in private local environments. When online tools are used, temporary or private modes may reduce data retention but do not replace confidentiality safeguards. AI suggestions must not be used as final assessment text without full rewriting and expert verification. Any text suggestions must be checked for factual accuracy, neutrality and originality. Representative prompts must be recorded in the DMR, along with their purpose, such as summarising, paraphrasing or harmonising tone. Prompts should be phrased neutrally to avoid introducing unintended interpretations or bias.

Coding usage

AI-powered coding and scripting assistants can support writing, debugging and optimising code used for data processing, figure generation or text analysis. These tools can improve efficiency but require careful validation. Experts must maintain full understanding of any code used in the assessment.

Coding and scripting support:

Usage: Supporting the writing and improving of code for data analysis, figure generation and text extraction.

Risks: Inaccurate or insecure code; accidental disclosure of confidential data, access tokens or API keys; reliance on opaque model logic; limited reproducibility.

Guidelines: Experts may use AI tools for code generation or debugging with caution. Confidential data, credentials and access tokens must never be shared with AI systems. All AI-generated code must be reviewed, tested and fully understood by the expert before use, including the logic and statistical assumptions embedded in the code. Exceptions for local or institutionally hosted tools may apply where confidentiality is ensured.

The use of AI-assisted coding must be documented in the DMR. Experts should also record representative prompts or instructions used for code generation or debugging, including their purpose, to support transparency and reproducibility. AI-generated code should not independently determine or select statistical models; methodological decisions must remain expert led.

Translational usage

AI translation tools can support inclusivity by facilitating multilingual participation and access to non-English materials. Their use must respect confidentiality, accuracy and traceability. AI translations of draft content must be used with caution and may only be processed through private or institutional systems that ensure confidentiality. Text processed through AI translation tools must be independently reviewed by experts fluent in the target language.

Translation of draft sections into English

Usage: Internal use for translation of draft materials.

Risks: Disclosure of confidential text; inaccurate or inconsistent translation; loss of nuance; dependency on unverifiable algorithms.

Guidelines: Experts may use private or institutional AI translation tools, preferably offline. Draft text should never be uploaded to public or third-party AI systems. Translations of primary sources of evidence should be processed in short sections. AI-translated sections should be clearly marked for internal review. All translations must be reviewed by experts fluent in English to ensure accuracy, consistency and adherence to assessment meaning. Representative prompts or translation settings, such as "translate to English for technical style", should be noted in the DMR. Experts are responsible for synthesizing content for the assessment coming from the translated source materials (e.g. reports, papers or grey literature).

Translation of evidence materials

Usage: Internal use for translating source materials such as national reports or grey literature.

Risks: Misinterpretation; loss of contextual meaning; stylistic drift; potential over-reliance on automated outputs.

Guidelines: Experts may use AI tools to translate evidence materials, but translated outputs must always be reviewed by a domain expert fluent in the original language. Translations should be used as an aid and not as a substitute for expert understanding. Any use of AI translation for evidence extraction should be documented in the DMR.

Translation of search terms

Usage: During the definition of literature search protocols.

Risks: Incorrect or misleading translations that alter search sensitivity or scope.

Guidelines: Experts may use AI tools to translate search terms, but all translations must be reviewed by a domain expert to ensure semantic accuracy and avoid misalignment in search strategies. Where AI is used, representative prompts and the resulting translated terms should be recorded in the DMR for transparency.

Literature search

AI can assist in expanding the scope of literature searches through the generation of search terms, identification of potentially relevant studies and summarisation of findings. However, AI tools must never replace systematic expert-led search methods. All AI-suggested references must be verified by experts for existence, accuracy and relevance.

Generation of search terms

Usage: Defining search protocols.

Risks: Overly broad or irrelevant results; missing key terms; model bias.

Guidelines: Experts may use AI tools to draft or expand search strings, provided results are refined and validated manually by domain experts. Representative prompts used to generate search terms should be documented in the DMR.

Search via general chatbots

Usage: Supplementing expert-based searches.

Risks: Fabricated citations; unverifiable sources; biased or incomplete reference lists; non transparent selection of references.

Guidelines: Experts must not rely on chatbots as primary search methods. They may use them only to explore topics or identify keywords, and all results must be independently checked. Any prompts used for exploratory searches must be documented in the DMR, including the intended scope of the query and representative wording, to support reproducibility and avoid misinterpretation.

Use of AI-based search tools

Usage: Initial screening of literature.

Risks: Incomplete database coverage; algorithmic bias; fabricated or irrelevant references; limited transparency regarding data sources.

Guidelines: Experts may combine AI-assisted search tools with established bibliographic databases such as OpenAlex, Scopus or Web of Science. All references identified through AI tools must be manually verified for existence and suitability. Limitations of AI search tools, including database coverage, filters and potential bias, should be documented in the DMR.

Literature and data analysis

AI can support large scale text or data analysis, but outputs must be critically validated. AI tools may assist in identifying patterns, summarising content or organising information, but they must not generate new scientific findings or independent interpretations.

PDF/Text analysis and synthesis

Usage: Extracting metadata, summarising results and categorising content.

Risks: Bias in model outputs; limited reproducibility; misclassification; fabricated; patterns; variation across model versions.

Guidelines: Experts may use AI tools to support text and PDF analysis, but all outputs must be cross checked using independent expert review or manual methods. AI model configurations and constraints should be recorded in the DMR. Prompts or instructions used to guide AI models in extracting, summarising or categorising text must be retained in the DMR together with representative outputs, supporting traceability of analytical steps. Where available, experts may refer to model cards or system cards describing the characteristics and limitations of the AI tools used.

Statistical analysis

Usage: Supporting hypothesis formulation or generating analysis code.

Risks: Inappropriate or non-transparent statistical methods; incorrect assumptions; model misspecification; fabricated outputs.

Guidelines: Experts must not use AI tools to autonomously perform statistical analyses, generate new results or interpret model outputs. AI may assist with drafting or improving code, but selection of methods, models and parameters must remain expert led. Any AI generated code must be reviewed, tested and fully understood before use. All decisions related to modelling and statistical interpretation must be made by experts based on established scientific practice. The use of AI and representative prompts must be documented in the DMR.

Figures

AI tools can help conceptualise or prototype figures but must never substitute expert verification, manual refinement or expert led data interpretation. All final figures included in IPBES assessments must be created and verified by experts to ensure accuracy, neutrality and reproducibility.

Conceptual figures

Usage: Drafting preliminary layouts or illustrations.

Risks: Copyright issues; biased or culturally insensitive imagery; inaccurate representations; lack of transparency in image generation.

Guidelines: Experts may use AI tools for early sketches or mock ups, but all final conceptual figures must be manually prepared and reviewed by experts. AI prompts used to generate or prototype figures, such as textual descriptions, layout suggestions or stylistic indications, should be included in the DMR. Experts should ensure that prompts do not contain copyrighted or culturally sensitive content and should verify that outputs do not introduce stereotypes or unintended interpretations.

Data figures

Usage: Generating or styling data visualisations.

Risks: Misleading scales; inappropriate styling; incorrect statistical assumptions; reduced reproducibility; opacity of data transformations.

Guidelines: Experts may use AI to prototype code for generating figures, but all final figures must be based on source data, rendered manually and fully validated by experts. AI generated visualisations should not be used directly in assessment drafts. Any use of AI to support data visualisation should be documented in the DMR, including representative prompts used to request coding assistance. Experts must ensure that final figures accurately reflect underlying data and adhere to IPBES standards for transparency and reproducibility.

Review of assessment drafts

AI must not be used for reviewing IPBES drafts, as this would risk breaching confidentiality, undermining the credibility of the review process and compromising trust in the Platform. Draft report text must not be uploaded to any public or third-party AI system under any circumstances.

AI support in internal and external reviews

Usage: Strongly discouraged and generally not permitted.

Risks: Breach of confidentiality; disclosure of text outside authorised channels; biased or inaccurate feedback; erosion of trust in the review process; confusion between expert judgment and AI generated comments.

Guidelines: IPBES experts and reviewers must not upload draft text, review comments or any confidential material to AI systems. All reviews must be conducted independently and confidentially by designated experts. AI tools must not be used to generate, structure or edit review comments. Where experts rely on local or institutionally hosted tools for basic editorial support, no draft text may be shared beyond secured environments, and expert judgment must remain fully independent. Review comments generated or influenced by AI systems must not be submitted.

Implementation, monitoring and reporting

The Code of Practice on the use of artificial intelligence applies to all new and ongoing IPBES assessments and other knowledge synthesis processes. The expert groups preparing IPBES deliverables are required to adhere to its provisions. Each assessment TSU is responsible for supporting experts to ensure that the use of AI within its assessment follows this Code and that all relevant details are documented in the corresponding data and knowledge management reports (DMRs).

Compliance with this Code will be monitored through the review of DMRs and related documentation by the TSU on data and knowledge management. The TSU will report to the task force on the degree of compliance and on any recurring issues, challenges or good practices identified. Verification may include checking whether AI-generated content has been incorporated without disclosure, including through the examination of text consistency or the use of available reverse engineering tools, without relying on any single detection method. The task force will provide technical guidance to assessment TSUs and experts and will support measures that promote the responsible and ethical use of AI in all IPBES activities. During external review processes, any use of AI must be explicitly documented, including representative prompts, model names and relevant configuration details.

The task force on data and knowledge management will include in its regular reports to the Bureau and the Multidisciplinary Expert Panel information on the implementation of this Code, including lessons learned and proposals for refinement. The Code of Practice is a living document and will be reviewed regularly, at least every two years, to reflect new developments in AI technologies, ethical standards and relevant United Nations system principles and guidelines. Revisions will be proposed by the task force and submitted to the Bureau and the Multidisciplinary Expert Panel for endorsement.

Any emerging or unexpected issues concerning the use of artificial intelligence that are not covered in this Code will be brought to the attention of the IPBES secretariat and the technical support unit on data and knowledge management, who will seek the guidance of the task force on data and knowledge management. Guidance provided through this process will be considered in future update processes of the Code.

The task force and the TSU on data and knowledge management, in cooperation with the secretariat and other task forces as relevant, will facilitate the development of training materials, guidance notes and awareness raising activities to support experts and TSUs in the ethical and responsible use of AI. These activities will encourage the sharing of experiences and good practices among assessments, strengthen a common understanding of AI use across IPBES and help ensure consistency in the application of this Code.

Glossary

This glossary complements existing definitions contained in IPBES documents, including the data and knowledge management policy, assessment glossaries and other relevant Platform decisions, such as the approach to recognising and working with Indigenous and local knowledge adopted in annex II to decision IPBES 5/1. Only additional or specific terms relevant to the Code of Practice on the use of artificial intelligence are defined below.

AI assisted workflow. Any step within the assessment process, such as literature search, data analysis, figure design or text editing, where AI tools are used. Each instance, including representative prompts and resulting outputs, should be transparently recorded in the relevant DMR.

Artificial intelligence. A collective term for computer based systems capable of performing tasks that normally require human intelligence, such as language processing, reasoning, pattern recognition or decision making. In the context of this Code, artificial intelligence refers specifically to systems that generate content, such as text, code or images, in response to user input. This includes large language models, image generation models and comparable tools.

Confidential content. Any unpublished draft material, personal data, review comments or internal discussion relating to IPBES assessments that must not be shared with external or online AI systems.

Generative AI. A type of artificial intelligence model that creates new outputs such as text, images, code or audio based on patterns learned from large volumes of data. Examples include large language models and image generation systems.

Generative output. The text, code, image or other content produced by a generative AI model in response to a prompt. Experts remain fully responsible for reviewing and validating such outputs.

Institutionally hosted AI systems. AI applications or models installed and executed on institutional infrastructure that is accessible only to members of the institution. These systems are preferred when working with confidential IPBES materials to ensure data security and compliance with confidentiality requirements.

Large language model. A type of AI model trained on extensive text corpora to generate and interpret natural language. Large language models can assist with summarisation, translation, coding and content generation, but their outputs are probabilistic and may contain factual or logical errors.

Locally hosted AI systems. AI applications or models executed on private or institutional infrastructure and accessible only to authorised users, often a single expert conducting an AI assisted analysis.

Model specification. A description of the AI system used, including the model name, version, provider and configuration parameters such as temperature or context length, and whether it was accessed online or locally. Such information should accompany AI related entries in DMRs to support transparency and reproducibility. For public chatbots and comparable systems, only the system name and model information may be available.

Online or third party AI systems. Commercial or public AI platforms accessed through the internet that process data externally. Their use in IPBES assessments requires explicit caution and disclosure and must not involve confidential content.

Prompt. A written instruction or query provided by the user to an AI system to generate a specific output. Representative prompts must be recorded in the DMR when AI tools are used for any step of the assessment process.